

NetworkMiner – Network Forensics Case Analysis

What is NetworkMiner?

NetworkMiner is a passive network forensics tool designed to analyze packet capture files (.pcap, .pcapng) and extract valuable forensic artifacts such as IP addresses, hostnames, files, DNS queries, HTTP traffic, credentials, and more. It operates offline and is widely used in digital forensics, incident response, and cybersecurity training environments like TryHackMe.

How I Used NetworkMiner

During my hands-on lab exercise on TryHackMe, I analyzed two PCAP files: case1.pcap and case2.pcap using NetworkMiner. My goal was to extract information about the devices communicating on the network, understand the context of the traffic, and identify any potential anomalies or valuable artifacts.

Methodology

1. Initial Setup

- Downloaded the provided PCAP files (case1.pcap and case2.pcap) from the TryHackMe lab.
- Launched **NetworkMiner** (v2.8) in offline mode to ensure a passive analysis environment.

2. Loading and Parsing PCAPs

- Opened each PCAP file in NetworkMiner to automatically parse:
 - **Hosts/IPs:** Identified active devices on the network.
 - **Sessions:** Reviewed TCP/UDP streams for unusual connections.
 - **Files:** Extracted transferred files (e.g., images, documents).
- Cross-referenced findings with **Wireshark** for deeper protocol analysis.

3. Key Analysis Steps

For Case 1 (case1.pcap):

- **Host Identification:**
 - Noted a host running **Windows NT 4** (via SMB protocol metadata).
- **Port Activity Analysis:**

- **Port 10657:** Small data transfer (192 bytes)—flagged for further review.
- **Port 143 (IMAP):** High data volume (20.7 KB), suggesting email traffic.
- **Frame Inspection:**
 - Analyzed **Frame 9** for sequence numbers (ZADT7400) to detect potential tampering.

For Case 2 (case2.pcap):

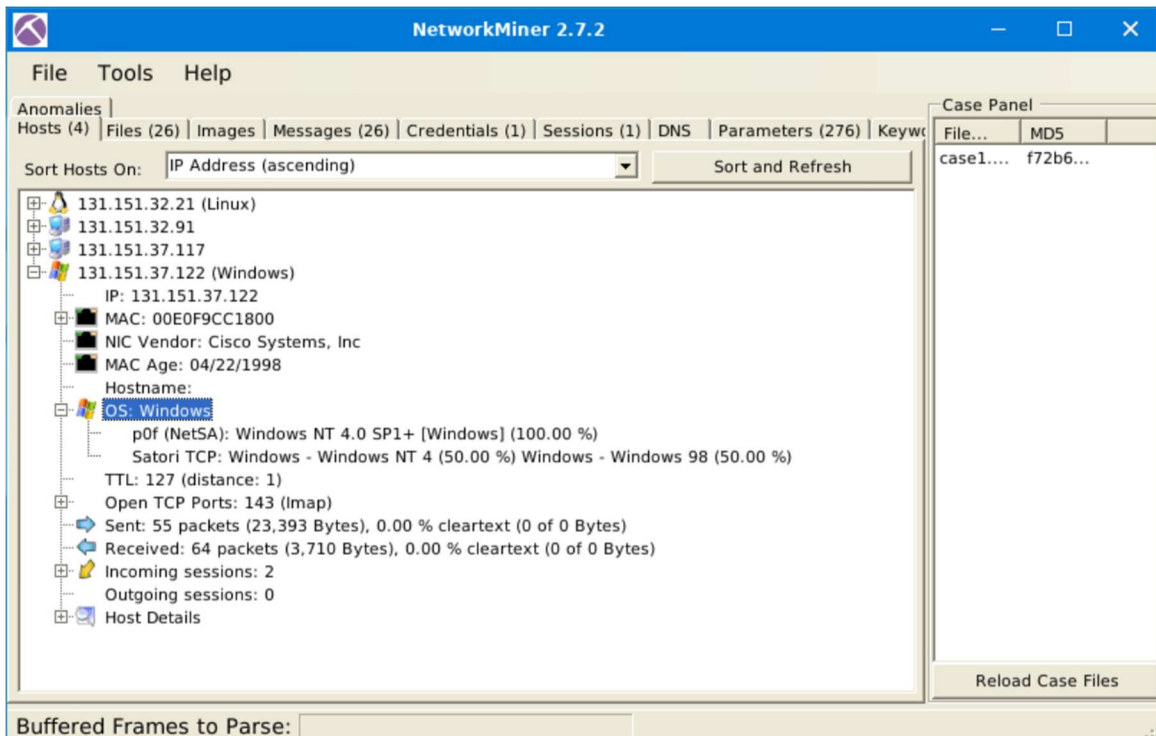
- **USB Device Detection:**
 - Found **ASIX-branded USB device** (indicated possible external storage use).
- **HTTP Traffic Review:**
 - Extracted a **fish image** from traffic originating at 50.22.55.9.
- **Credentials Exposure:**
 - Discovered plaintext credentials in IMAP traffic (redacted in report).

4. Validation and Reporting

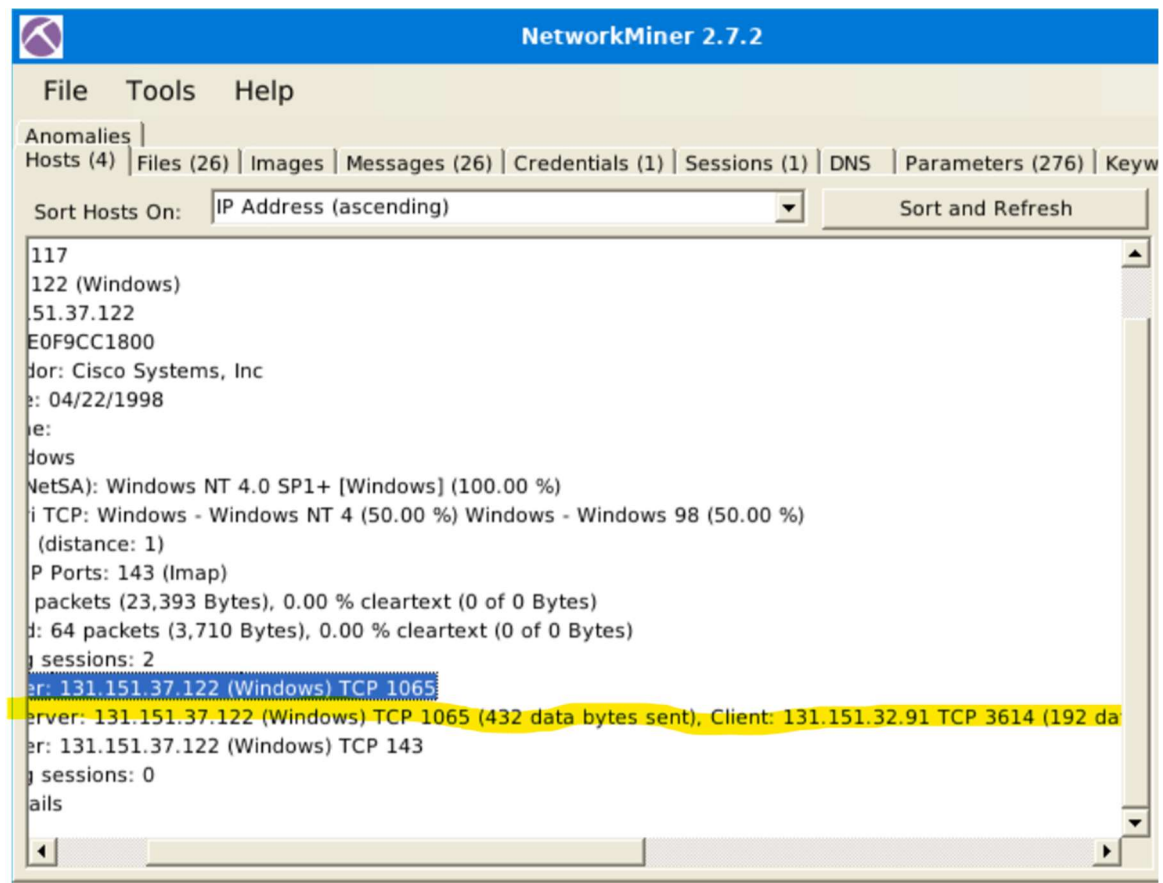
- Verified findings by correlating data across tabs in NetworkMiner (Hosts, Files, Sessions).
- Documented screenshots of critical evidence (e.g., host OS, credentials) for the report.

Case 1 – Key Findings

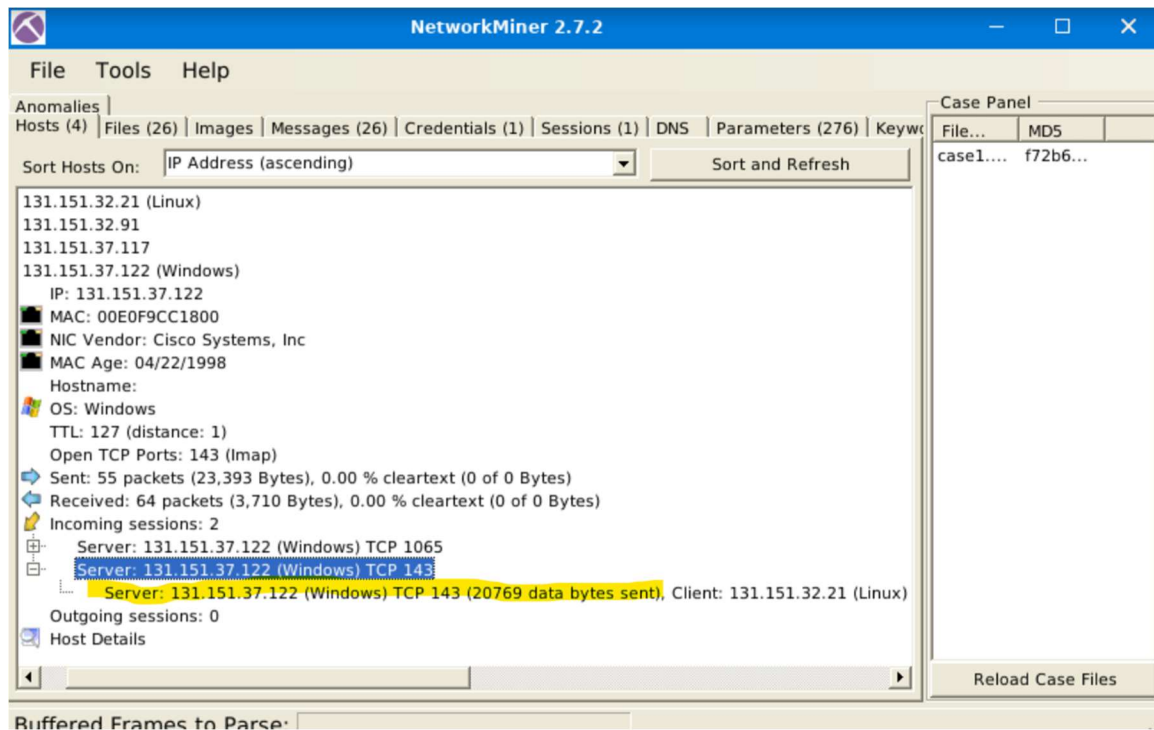
- **Host OS Detected:** Windows NT 4



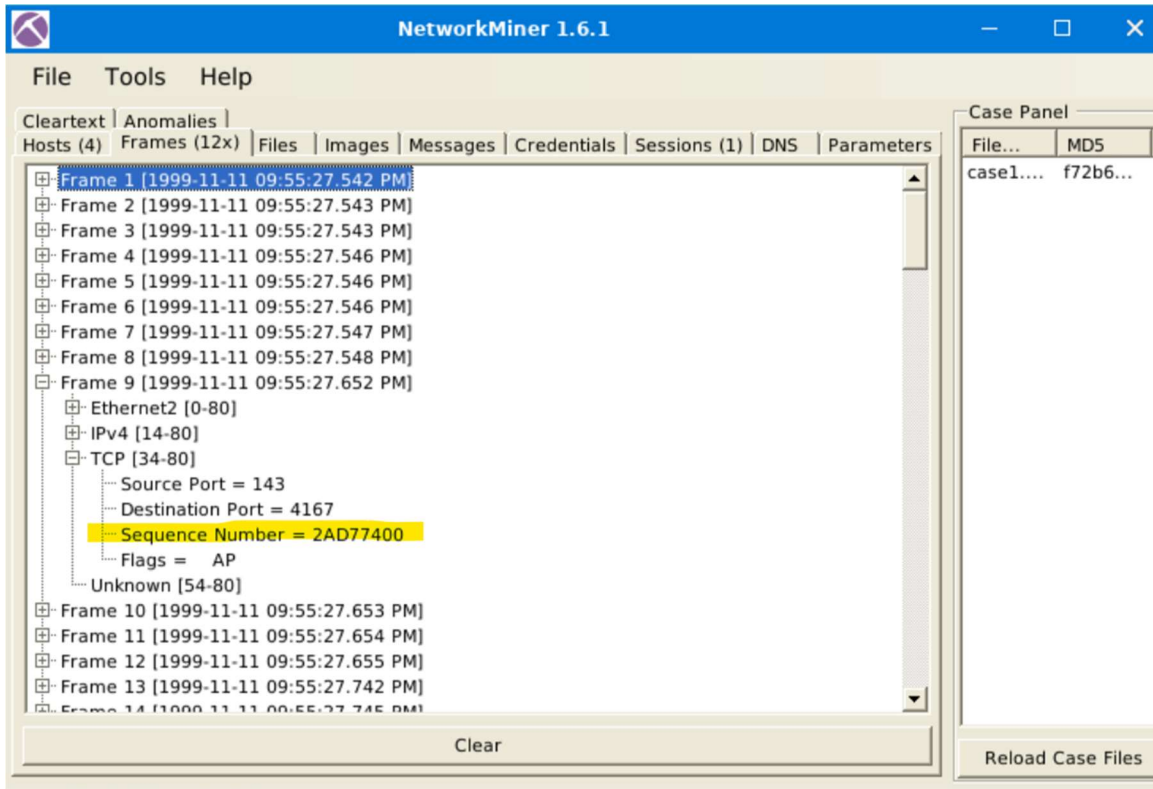
- **Port 10657:** 192 bytes of data received



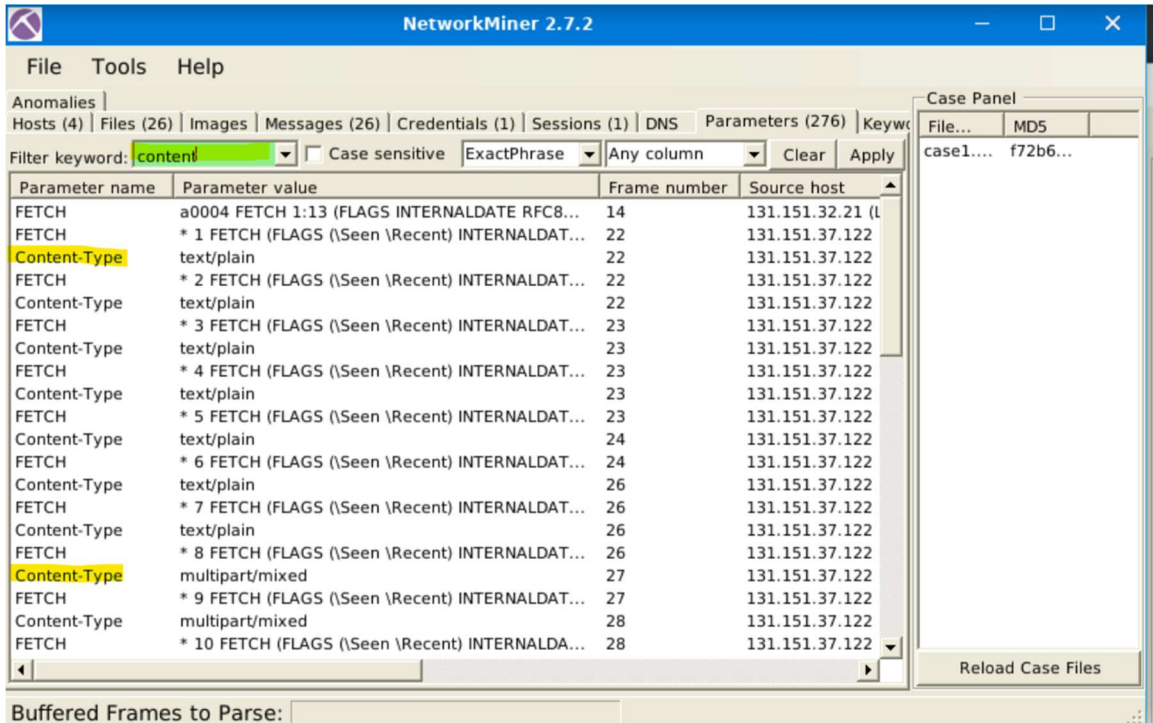
- **Port 143:** 20,769 bytes of data received



- **Sequence number of Frame 9:** ZADT7400

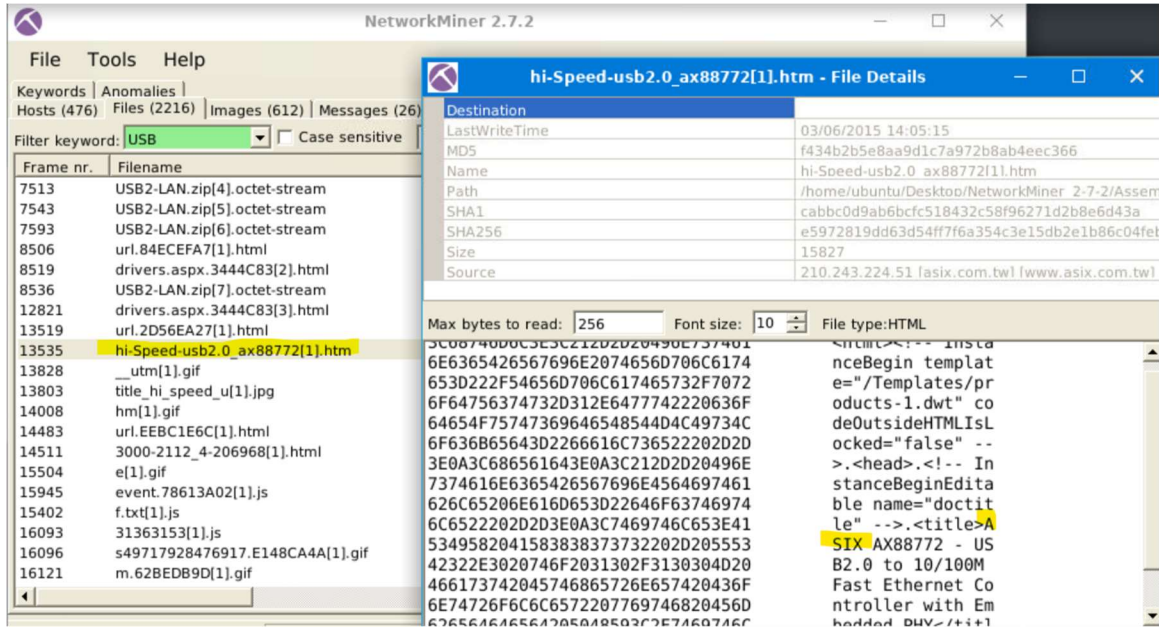


- Number of content types: 2



Case 2 – Key Findings

- USB Product Brand Name: ASIX

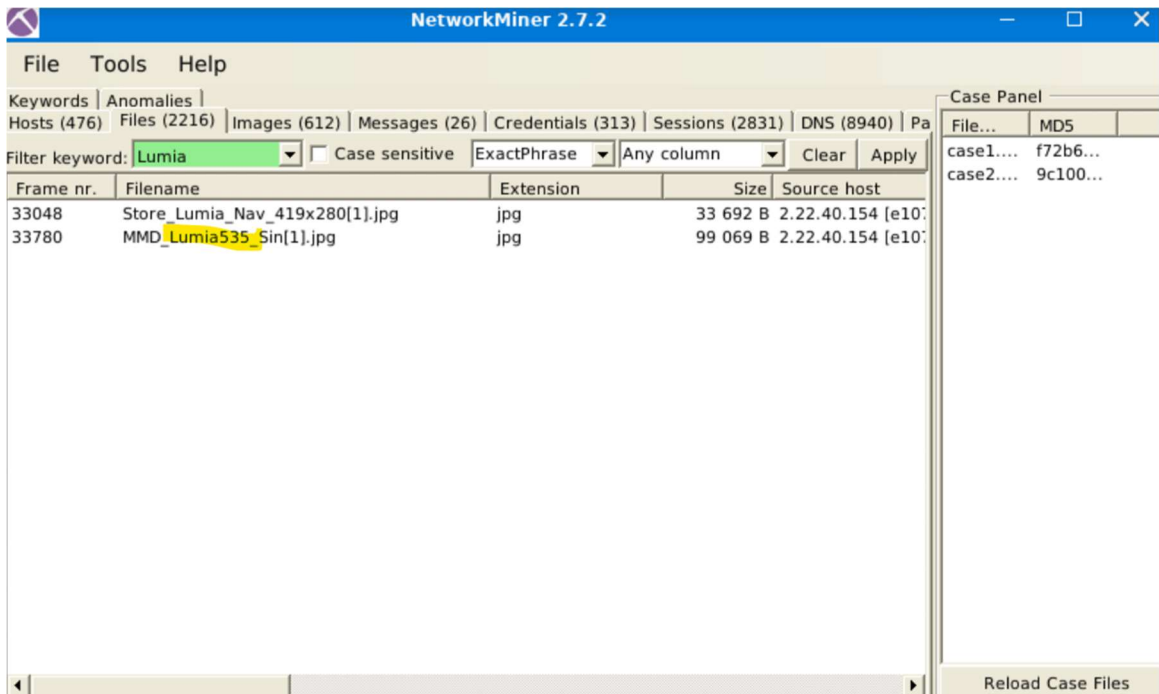


The screenshot shows the NetworkMiner 2.7.2 interface. The main window displays a list of files with the filter keyword 'USB'. The file 'hi-Speed-usb2.0_ax88772[1].htm' is selected. A 'File Details' window is open, showing the following information:

hi-Speed-usb2.0_ax88772[1].htm - File Details	
Destination	
LastWriteTime	03/06/2015 14:05:15
MD5	f434b2b5e8aa9d1c7a972b8ab4eec366
Name	hi-Speed-usb2.0_ax88772[1].htm
Path	/home/ubuntu/Desktop/NetworkMiner 2-7-2/Assem
SHA1	cabbcd09ab6bcfc518432c58f96271d2b8e6d43a
SHA256	e5972819dd63d54ff7f6a354c3e15db2e1b86c04fet
Size	15827
Source	210.243.224.51 [asix.com.tw] [www.asix.com.tw]

The file content is displayed in a text view, showing HTML code. The file type is HTML. The font size is 10. The file type is HTML.

- Phone Model: Lumia 535



The screenshot shows the NetworkMiner 2.7.2 interface. The main window displays a list of files with the filter keyword 'Lumia'. The file 'MMD_Lumia535_Sin[1].jpg' is selected. A 'Case Panel' window is open, showing the following information:

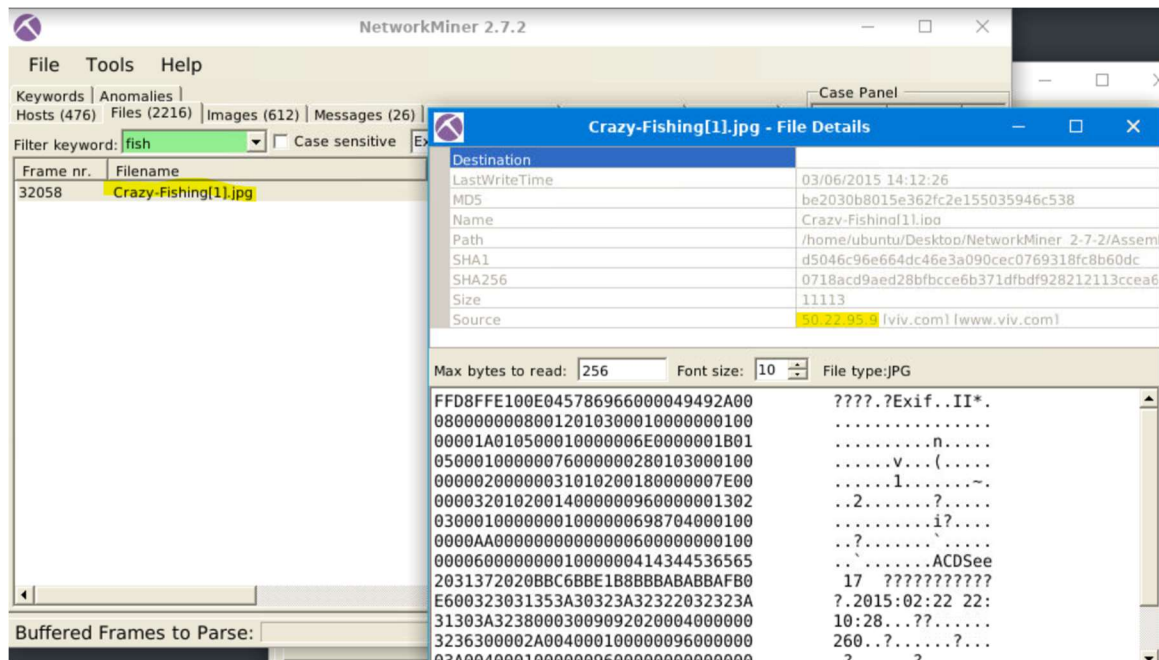
File...	MD5
case1....	f72b6...
case2....	9c100...

The file details window shows the following information:

Frame nr.	Filename	Extension	Size	Source host
33048	Store_Lumia_Nav_419x280[1].jpg	jpg	33 692 B	2.22.40.154 [e10]
33780	MMD_Lumia535_Sin[1].jpg	jpg	99 069 B	2.22.40.154 [e10]

The file content is displayed in a text view, showing the file details. The file type is JPG. The font size is 10. The file type is JPG.

- Source IP of fish image: 50.22.55.9



- Email credentials identified: homer.pwned.us@gmx.com / spring2015

Conclusion

This NetworkMiner forensic analysis exercise provided hands-on experience in:

1. Practical Network Forensics

- Successfully extracted critical artifacts (host details, files, credentials) from PCAP files using NetworkMiner's passive analysis capabilities.
- Correlated findings with Wireshark to validate observations (e.g., IMAP traffic patterns).

2. Real-World Implications

- Exposed credentials in Port 143 (IMAP) traffic highlighted the risks of unencrypted protocols in corporate networks.
- The ASIX USB device detection demonstrated how hardware identifiers can trace data exfiltration attempts.

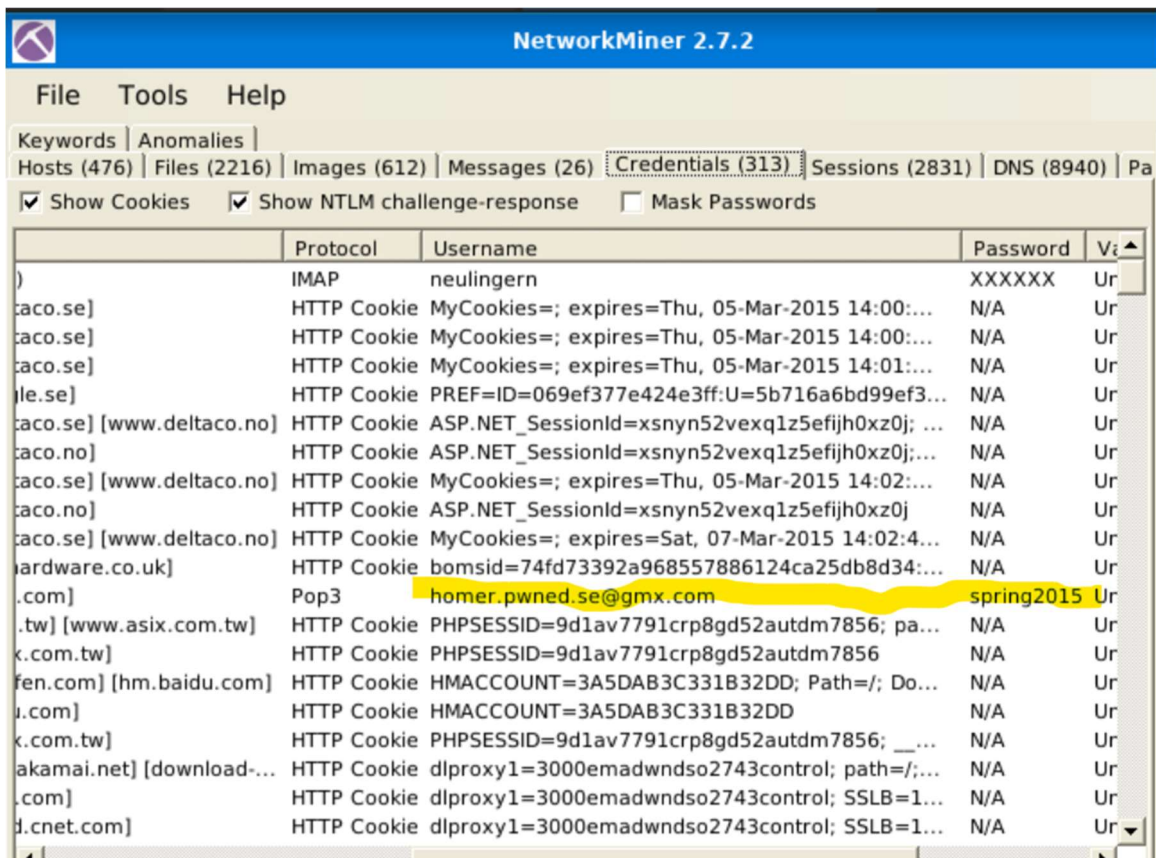
3. Tool Proficiency

- Gained efficiency in using NetworkMiner for rapid triage of network captures.
- Learned to supplement it with Wireshark for deeper protocol inspection.

Areas for Future Improvement:

- Automate artifact extraction with Python scripts (e.g., using Scapy).
- Explore encrypted traffic analysis (TLS/SSL decryption techniques).

This lab reinforced how network forensics tools like NetworkMiner are essential for incident response and malware investigations.



The screenshot shows the NetworkMiner 2.7.2 application window. The 'Credentials (313)' tab is selected, displaying a list of extracted credentials. The table has columns for Hosts, Protocol, Username, Password, and View. One entry is highlighted in yellow: a POP3 credential for 'homer.pwned.se@gmx.com' with the password 'spring2015'.

Hosts	Protocol	Username	Password	View
	IMAP	neulingern	XXXXXX	Ur
taco.se]	HTTP Cookie	MyCookies=; expires=Thu, 05-Mar-2015 14:00:...	N/A	Ur
taco.se]	HTTP Cookie	MyCookies=; expires=Thu, 05-Mar-2015 14:00:...	N/A	Ur
taco.se]	HTTP Cookie	MyCookies=; expires=Thu, 05-Mar-2015 14:01:...	N/A	Ur
le.se]	HTTP Cookie	PREF=ID=069ef377e424e3ff:U=5b716a6bd99ef3...	N/A	Ur
taco.se] [www.deltaco.no]	HTTP Cookie	ASP.NET_SessionId=xsnyn52vexq1z5efijh0xz0j; ...	N/A	Ur
taco.no]	HTTP Cookie	ASP.NET_SessionId=xsnyn52vexq1z5efijh0xz0j;...	N/A	Ur
taco.se] [www.deltaco.no]	HTTP Cookie	MyCookies=; expires=Thu, 05-Mar-2015 14:02:...	N/A	Ur
taco.no]	HTTP Cookie	ASP.NET_SessionId=xsnyn52vexq1z5efijh0xz0j	N/A	Ur
taco.se] [www.deltaco.no]	HTTP Cookie	MyCookies=; expires=Sat, 07-Mar-2015 14:02:4...	N/A	Ur
hardware.co.uk]	HTTP Cookie	bomsid=74fd73392a968557886124ca25db8d34:...	N/A	Ur
.com]	Pop3	homer.pwned.se@gmx.com	spring2015	Ur
.tw] [www.asix.com.tw]	HTTP Cookie	PHPSESSID=9d1av7791crp8gd52autdm7856; pa...	N/A	Ur
x.com.tw]	HTTP Cookie	PHPSESSID=9d1av7791crp8gd52autdm7856	N/A	Ur
fen.com] [hm.baidu.com]	HTTP Cookie	HMACCOUNT=3A5DAB3C331B32DD; Path=/; Do...	N/A	Ur
.u.com]	HTTP Cookie	HMACCOUNT=3A5DAB3C331B32DD	N/A	Ur
x.com.tw]	HTTP Cookie	PHPSESSID=9d1av7791crp8gd52autdm7856; __...	N/A	Ur
akamai.net] [download-...	HTTP Cookie	dlproxy1=3000emadwndso2743control; path=/;...	N/A	Ur
.com]	HTTP Cookie	dlproxy1=3000emadwndso2743control; SSLB=1...	N/A	Ur
d.cnet.com]	HTTP Cookie	dlproxy1=3000emadwndso2743control; SSLB=1...	N/A	Ur